

Finite Element Simulations in Materials

Modal Analysis of High-Performance Drone Motor Arm

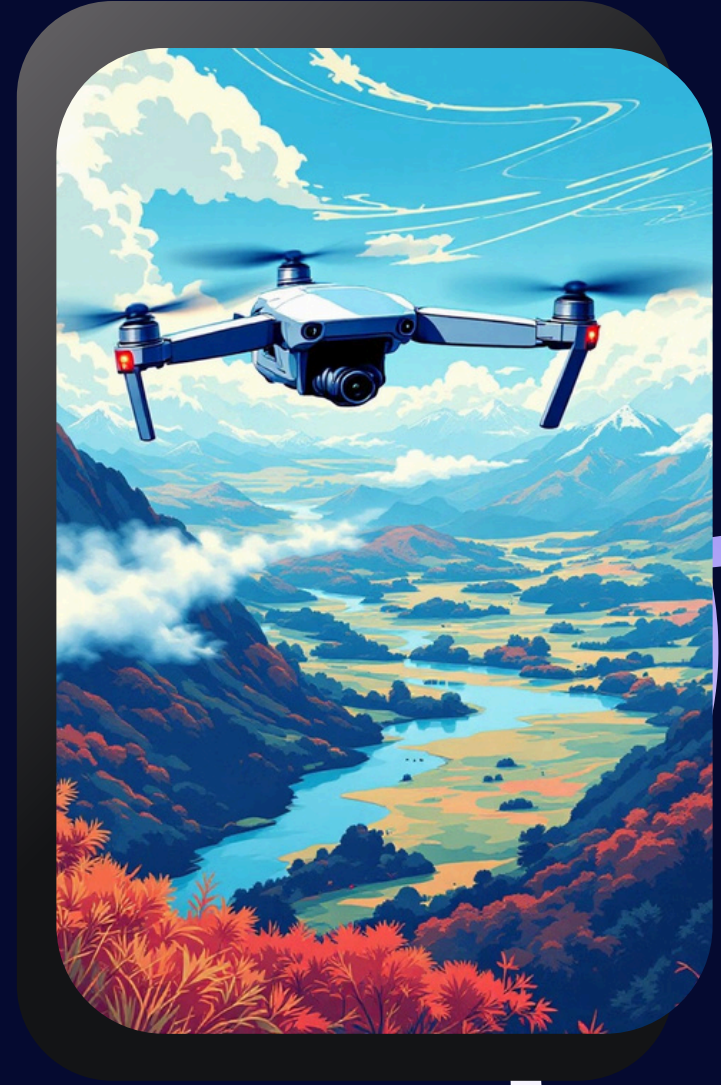
Software: Ansys Mechanical Workbench -2025 R2

By: Krushna More
Lakshya Rishi

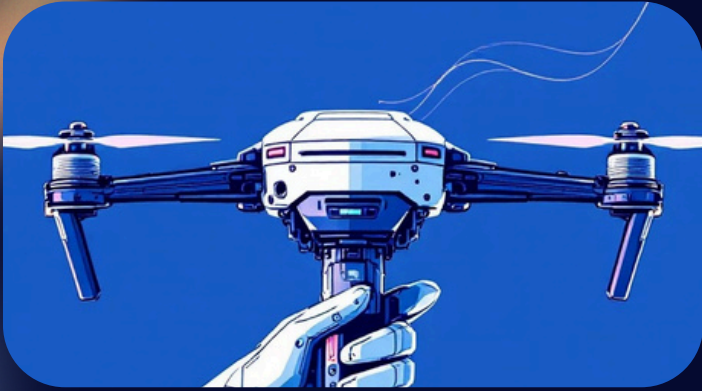
Problem Statement :

Modal Analysis: Preventing Drone Failure

Understanding vibration is crucial in aerospace and automotive engineering. Components often fail not from excessive load, but from resonance where vibrations amplify catastrophically."



The Resonance Problem: A Drone Motor Mount Case Study



- A Modal (Vibration) Analysis was conducted on a proposed drone arm design to evaluate structural safety at operating speeds.
- The objective was to determine if the motor's operating speed of 12,000 RPM would trigger destructive resonance.

Problem Statement & Objective

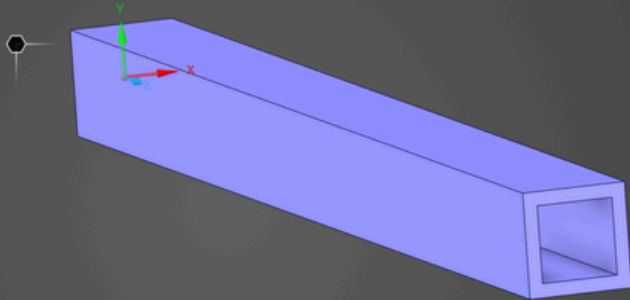
High-speed motors generate forced vibrations. If the frequency of these vibrations matches the natural frequency of the structural arm (Resonance), catastrophic failure can occur due to infinite displacement amplification.

Objective:

Identify the first 6 natural frequencies (modes) of the arm and ensure none fall within the 180-220Hz danger zone.



Tacoma bridge collapse , Washington



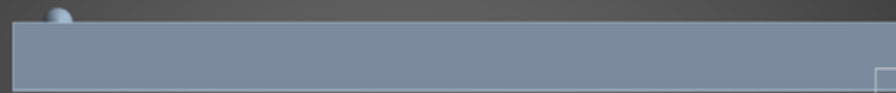
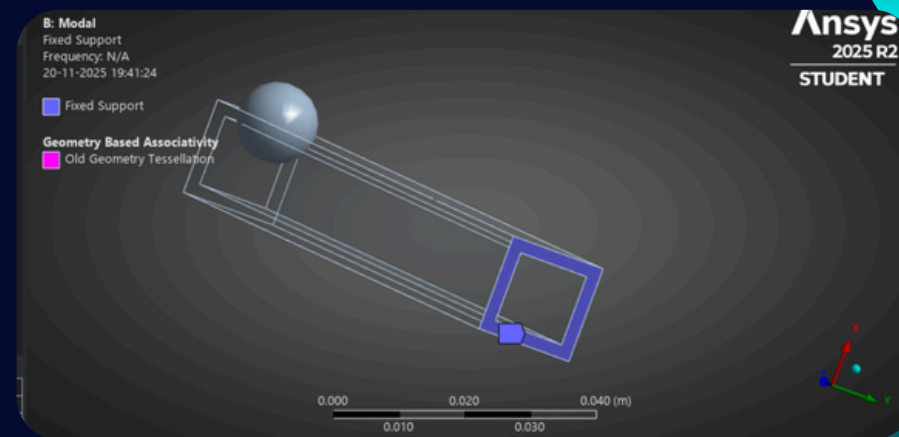
Geometry Definition

- Component: Cantilevered Arm (Square Tube Profile).
- Dimensions: Length 200mm , Cross-section 15mm x 15mm ; Thickness-2mm.
- Material: aluminium alloy (Default Ansys Library Material).

Loads & Boundary Conditions

- Support: Fixed Support applied to the base face.
- Loading: A Point Mass of 0.5kg applied to the tip face (simulating the motor weight and inertia)

This technique adds the motor's inertia without unnecessary mesh complexity, a crucial step for efficient simulation.



INPUT DATA & MATERIAL PROPERTIES

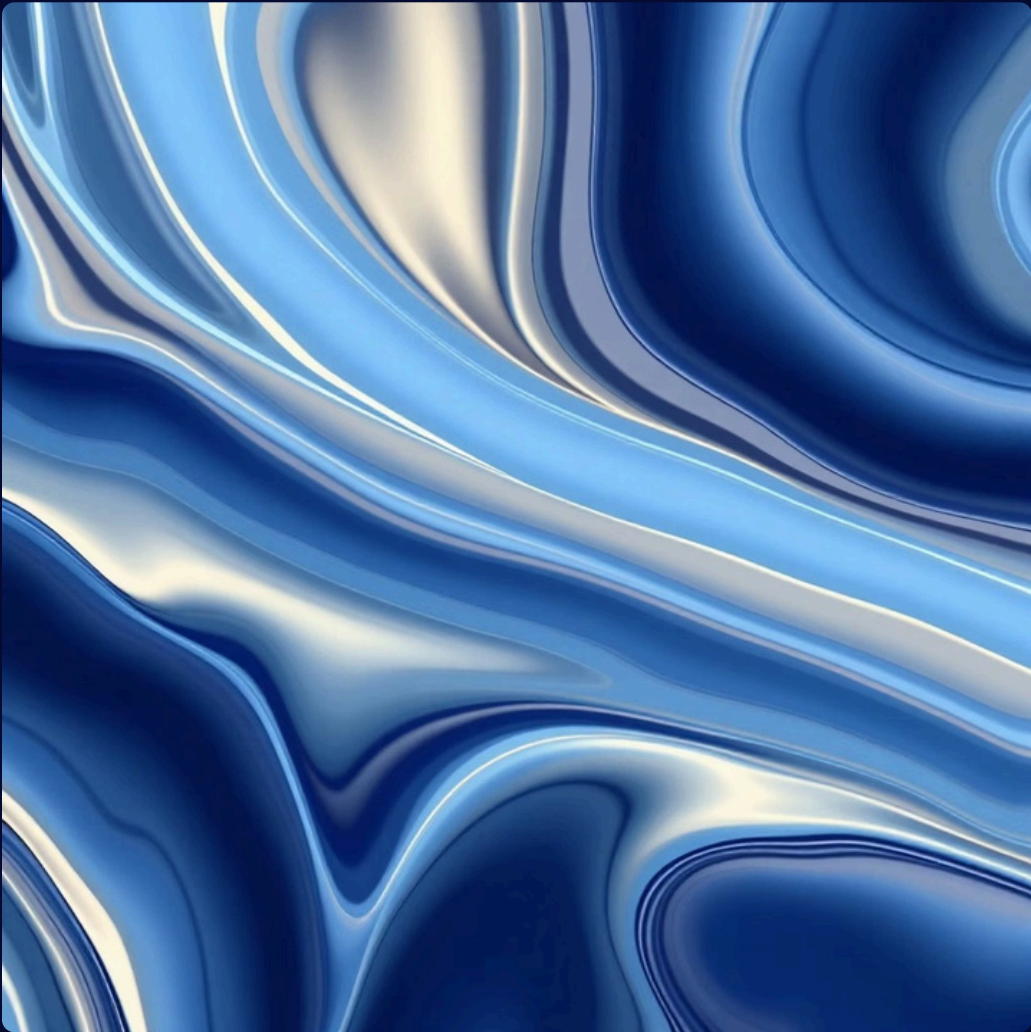
Material: Aluminum Alloy

Geometry: Hollow Square Tube

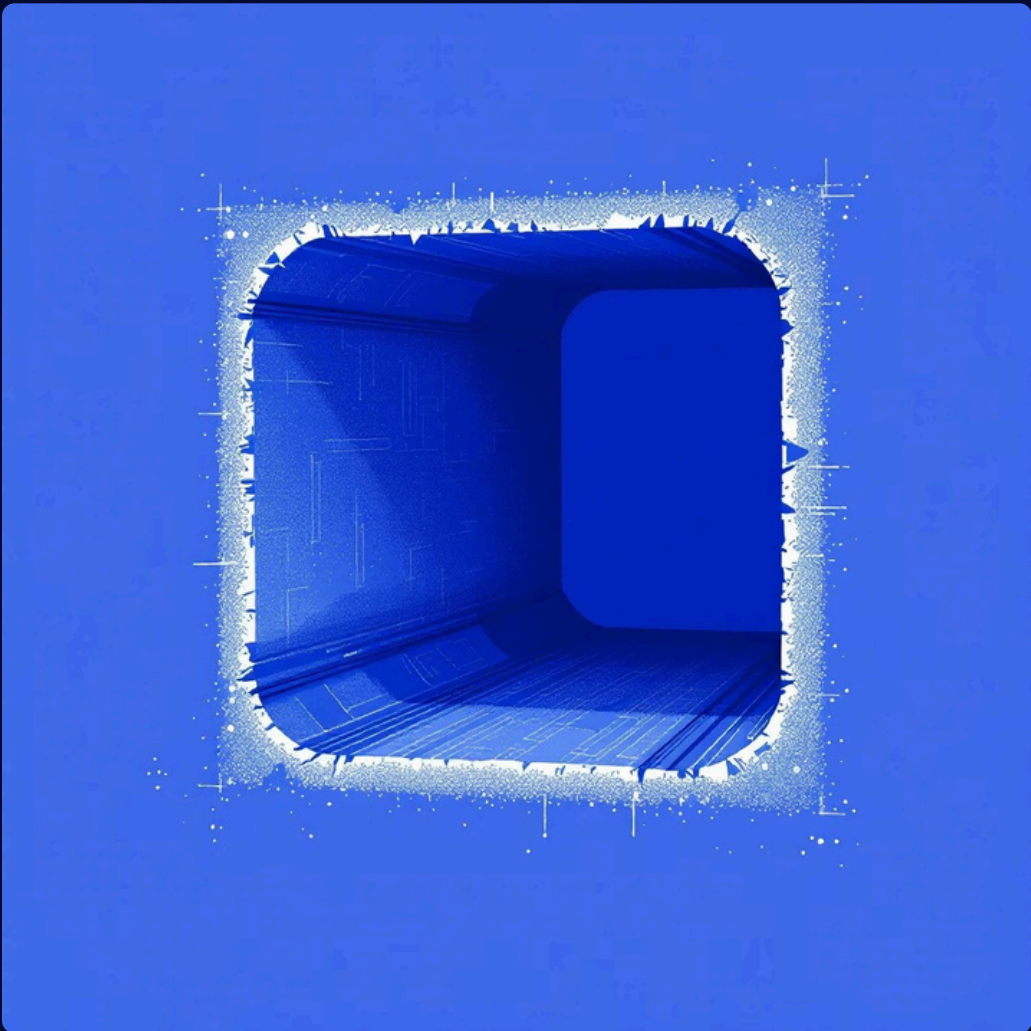
Length:200 mm
Cross-section: 15 mm x 15 mm Thickness: 2 mm

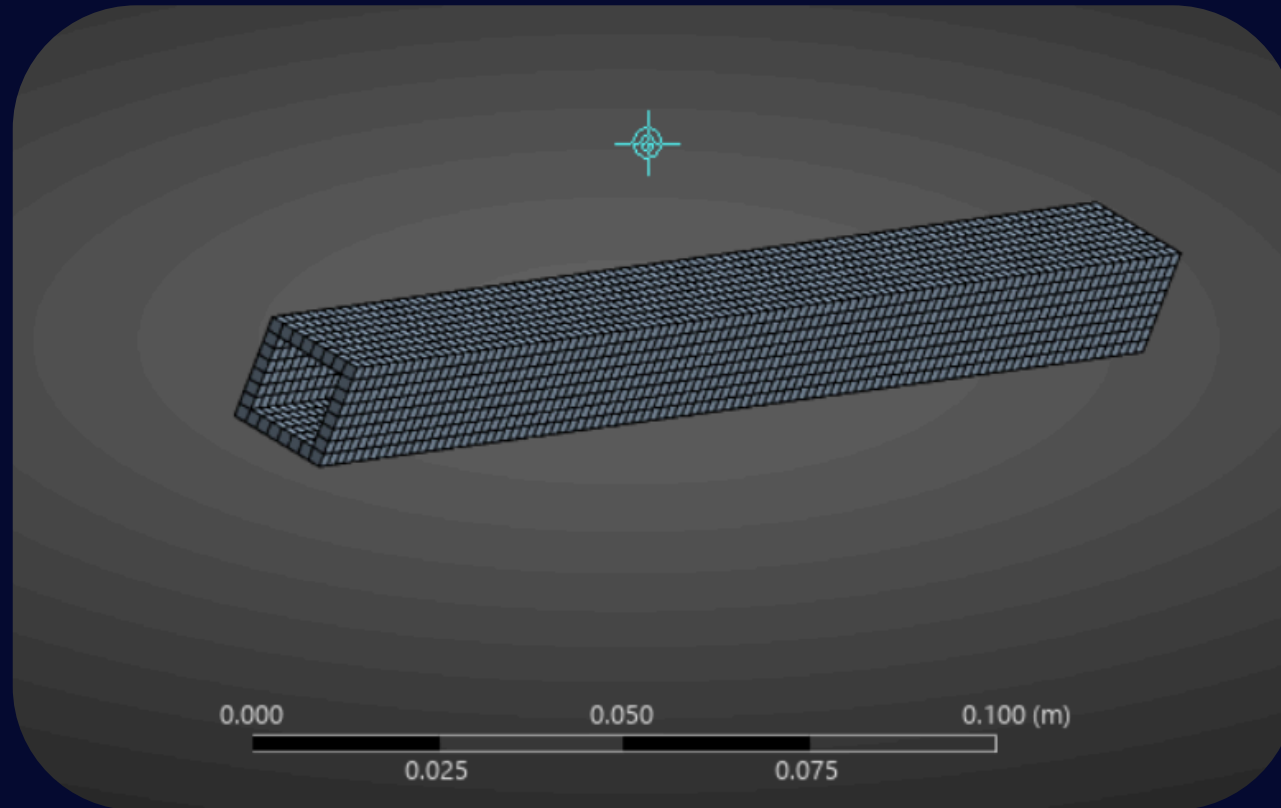
Motor Mass

0.05kg(50grams)placed at the arm's tip.



Material:





Meshing details

Apply a standard mesh (e.g., 2mm element size). Vibration analysis demands a fine mesh for accuracy.

no. of nodes : 19740

no. of elements: 2800

High-speed motors generate forced vibrations. If the frequency of these vibrations matches the natural frequency of the structural arm (Resonance), catastrophic failure can occur due to infinite displacement amplification.

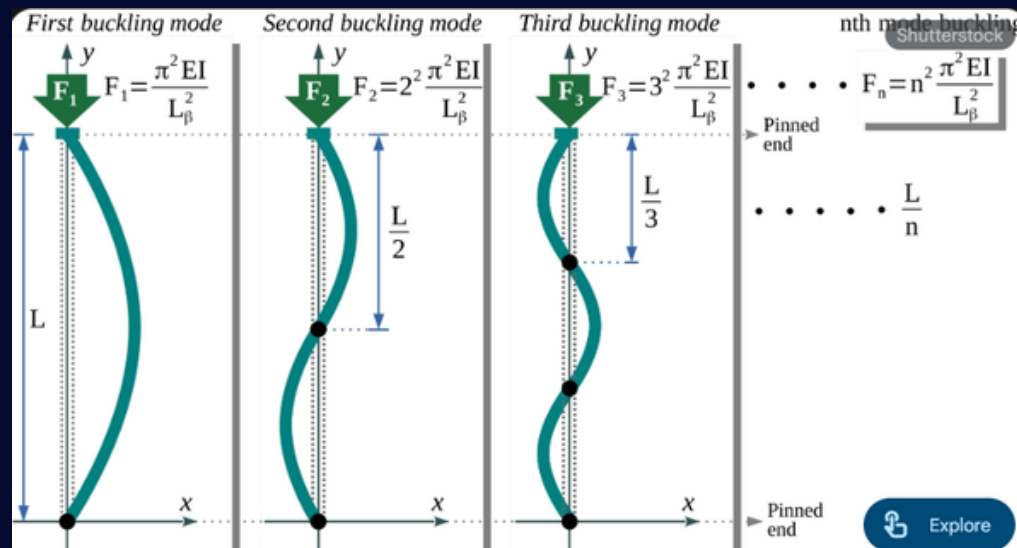
Operational Data:

- **Max Motor Speed:** 12,000 RPM
- **Excitation Frequency:** 200 Hz
- **Safety Margin Required:** $\pm 10\%$ (180 Hz – 220 Hz)

The forcing frequency (f) caused by the motor is calculated as:

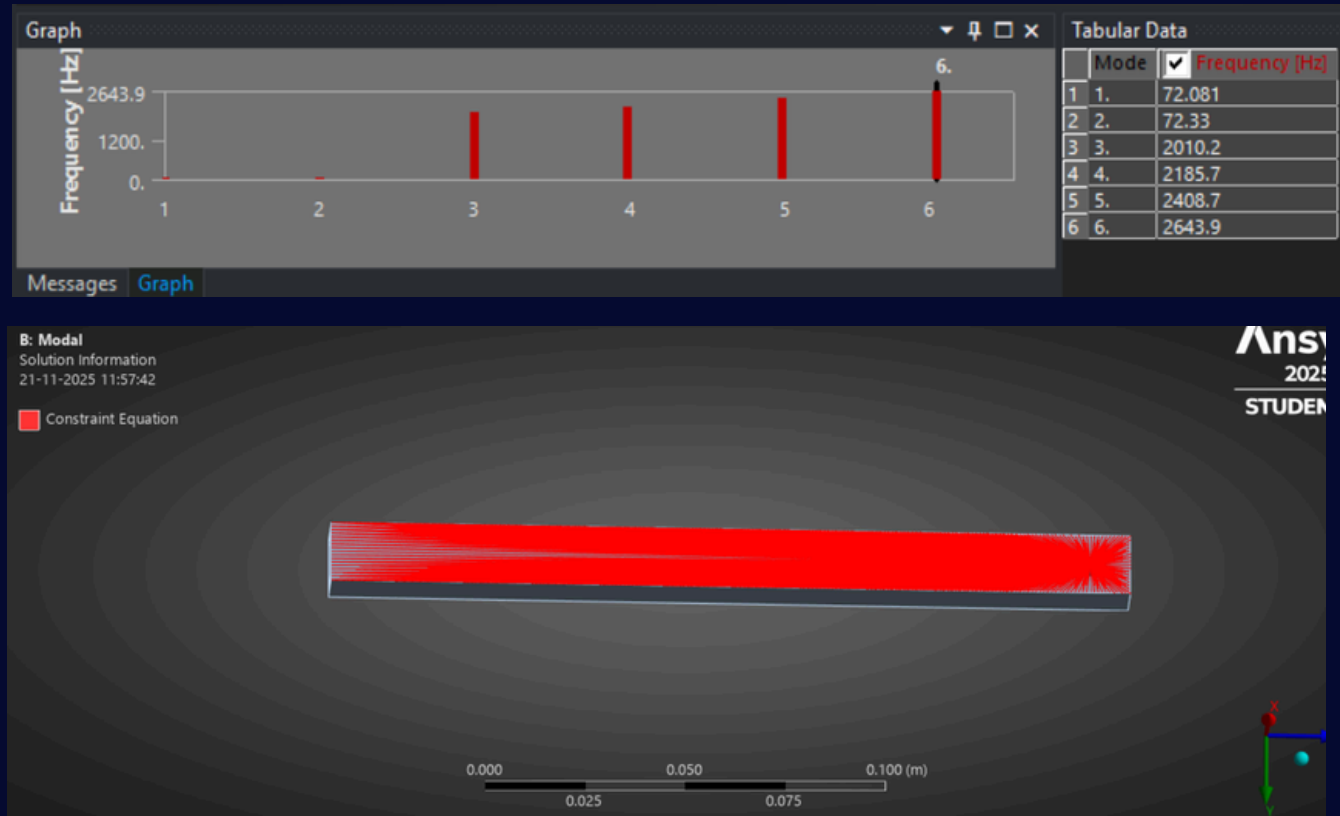
$$f = \frac{\text{RPM}}{60}$$

$$f = \frac{12,000}{60} = 200 \text{ Hz}$$



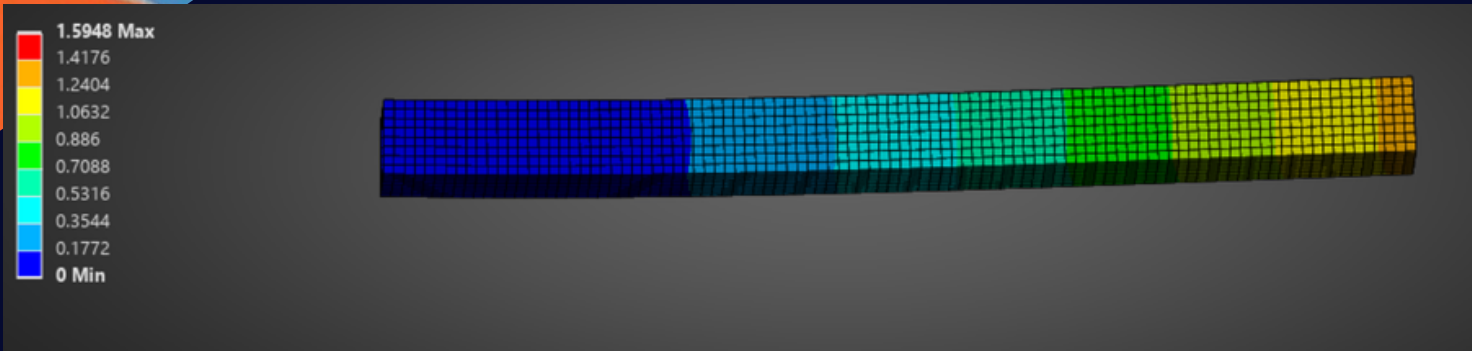
Go to Analysis Settings. Max Modes to to 6.
This typically captures the most critical vibrational behaviors (e.g., up/down, left/right, twist modes).

SIMULATION:



We compare the structural natural frequencies against the excitation frequency (200 Hz)

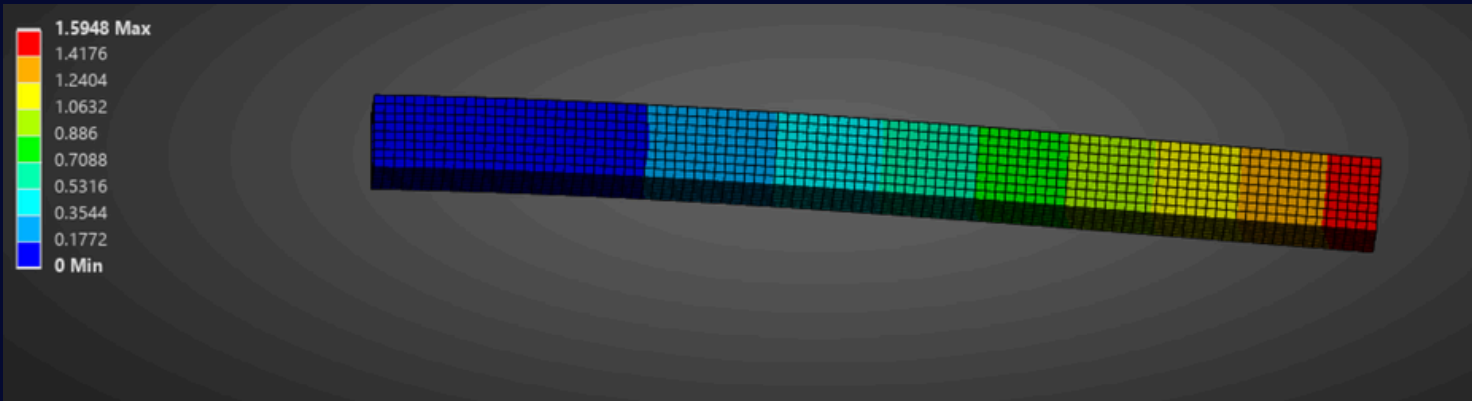
Check against Mode 1



**the first frequency (mode 1):
72.081 Hz**

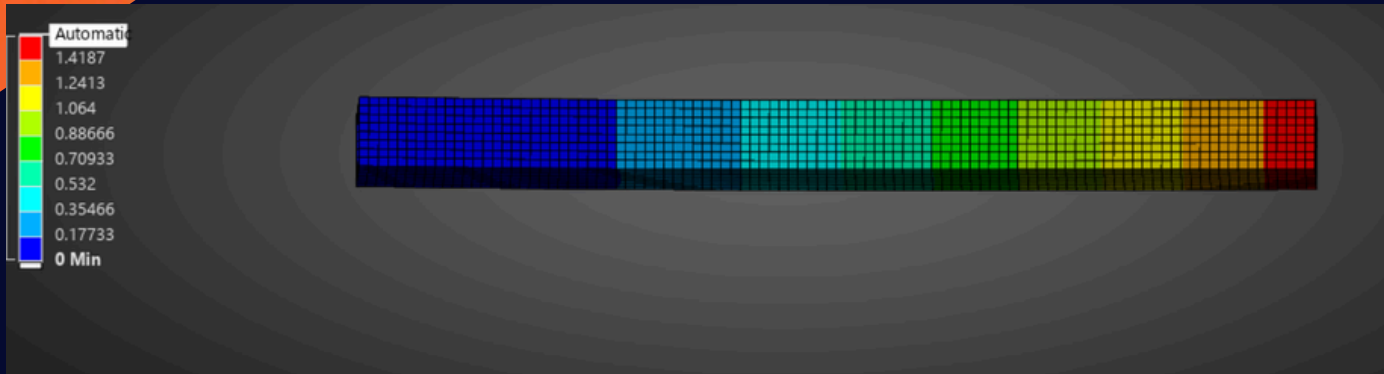
The motor spins at 200 Hz.

The motor spins significantly faster than the first natural frequency. This means the arm will pass through resonance briefly during "spin up" (when the motor accelerates from 0 to 12,000 RPM), but will be safe at top speed.



We compare the structural natural frequencies against the excitation frequency (200 Hz)

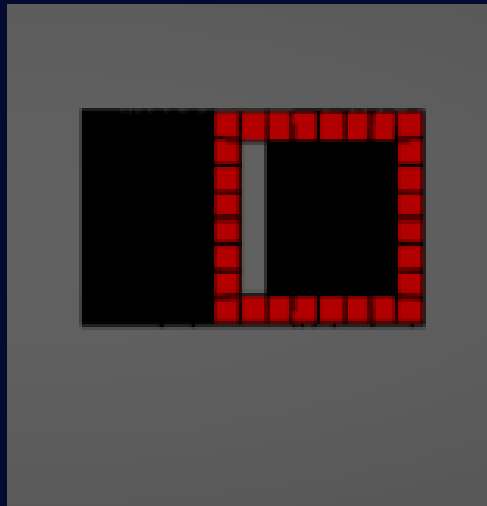
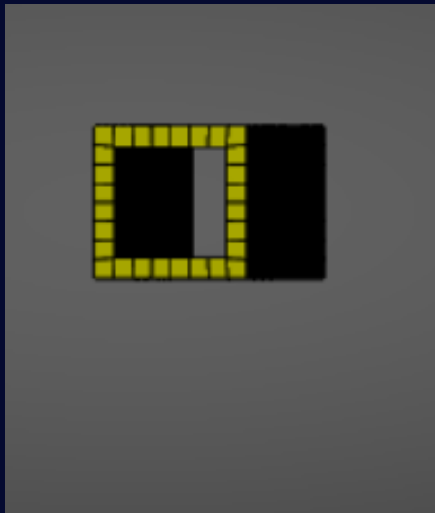
Mode 2



**the first frequency (mode 2):
72.33 Hz**

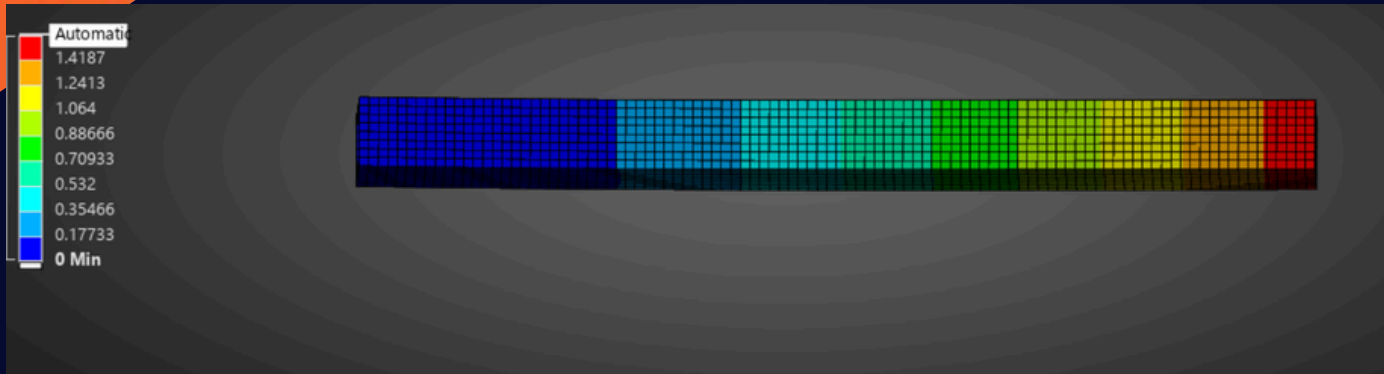
The motor spins at 200 Hz.

The motor spins significantly faster than the first natural frequency. This means the arm will pass through resonance briefly during "spin up" (when the motor accelerates from 0 to 12,000 RPM), but will be safe at top speed.



We compare the structural natural frequencies against the excitation frequency (200 Hz)

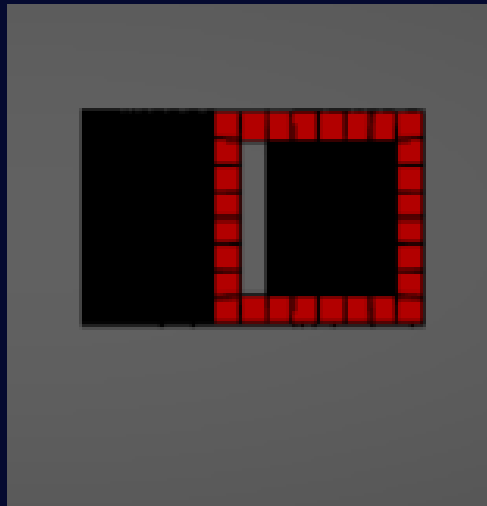
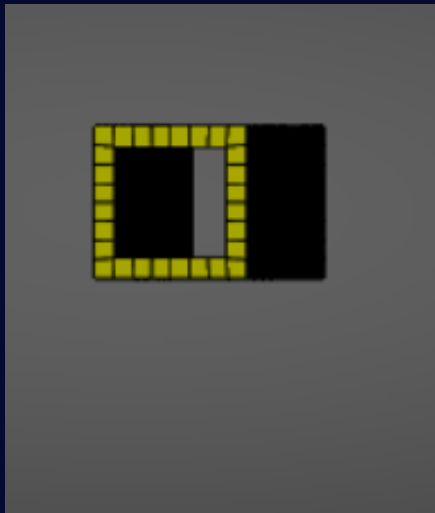
Mode 2



**the first frequency (mode 2):
72.33 Hz**

The motor spins at 200 Hz.

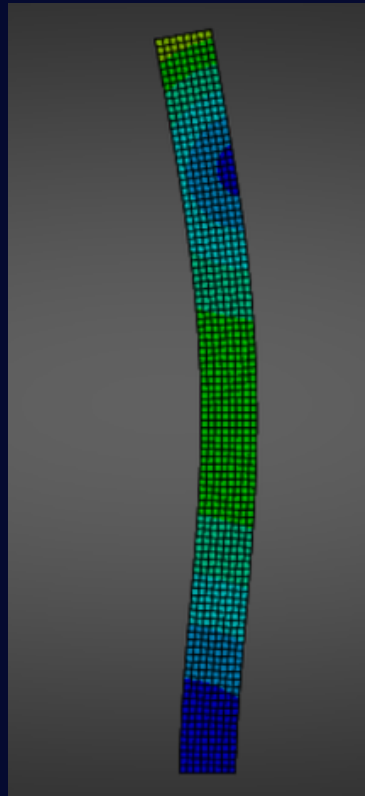
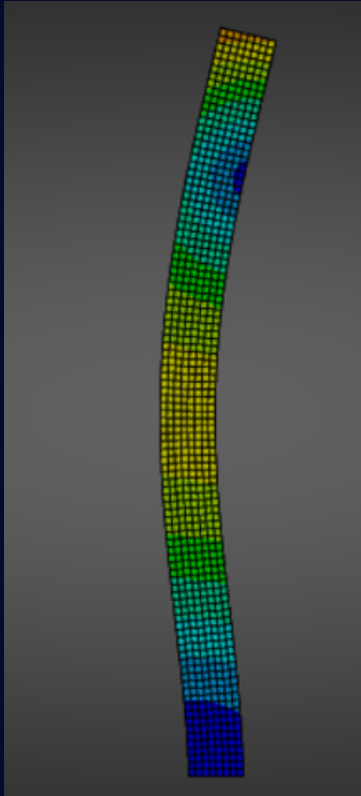
The motor spins significantly faster than the first natural frequency. This means the arm will pass through resonance briefly during "spin up" (when the motor accelerates from 0 to 12,000 RPM), but will be safe at top speed.



We compare the structural natural frequencies against the excitation frequency (200 Hz)

Mode 3

2010.2Hz



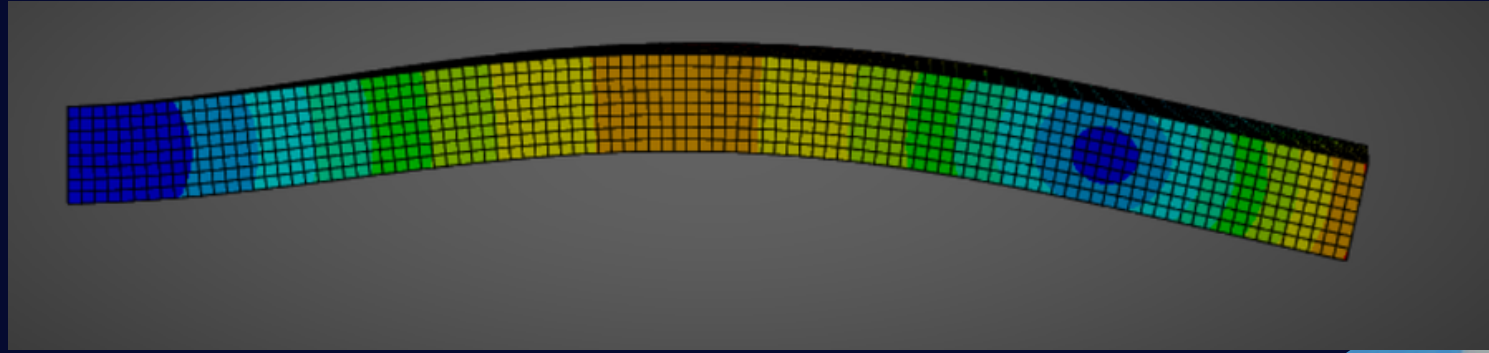
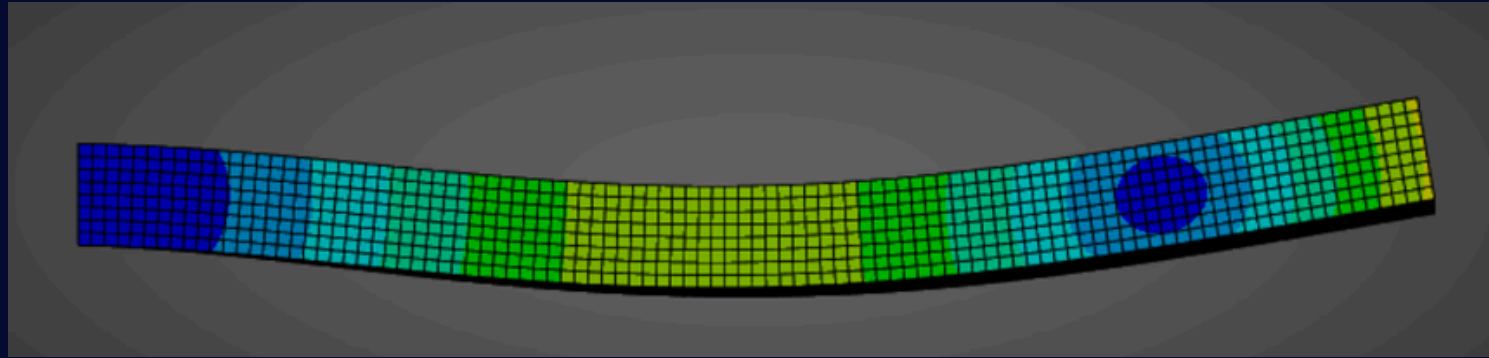
Check against Mode 3 (2010 Hz):

- The next vibration mode is at 2010 Hz.
- Status: This is extremely far away from the 200 Hz motor speed. There is zero risk of exciting this mode.

We compare the structural natural frequencies against the excitation frequency (200 Hz)

Mode 4

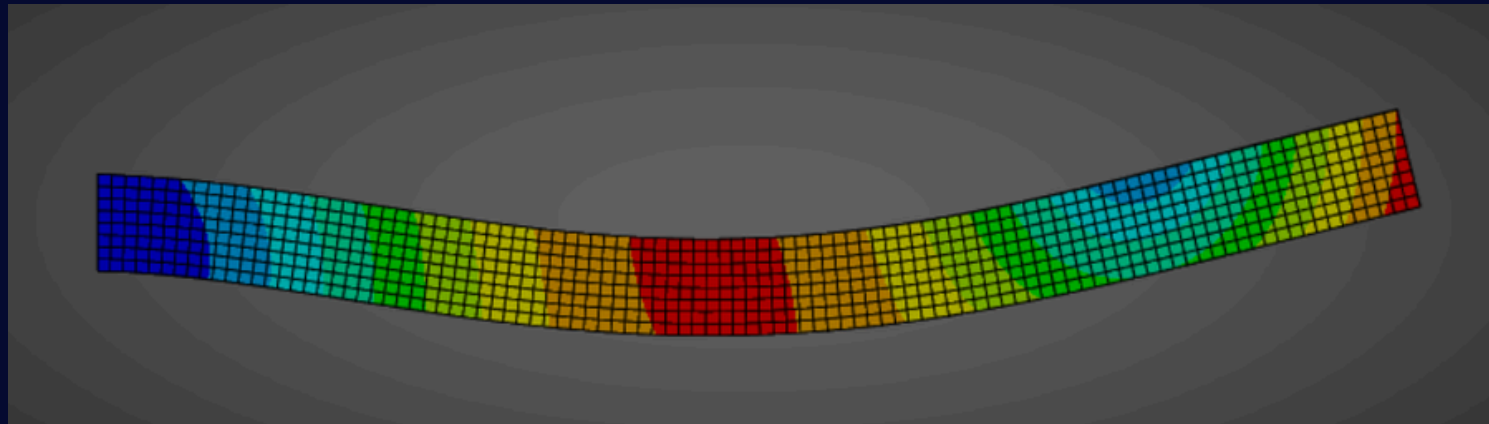
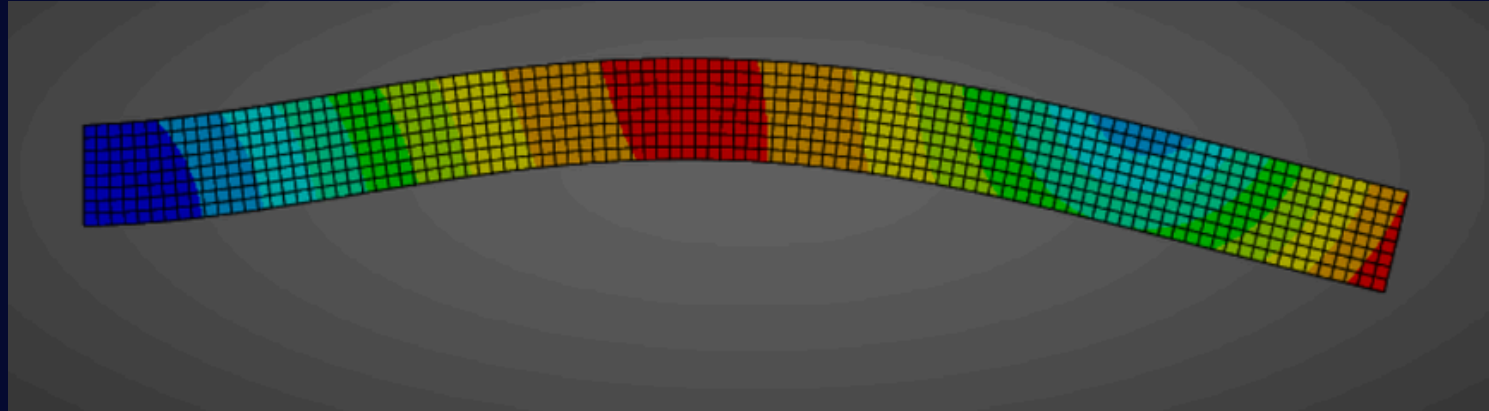
2185.7 Hz



We compare the structural natural frequencies against the excitation frequency (200 Hz)

Mode 5

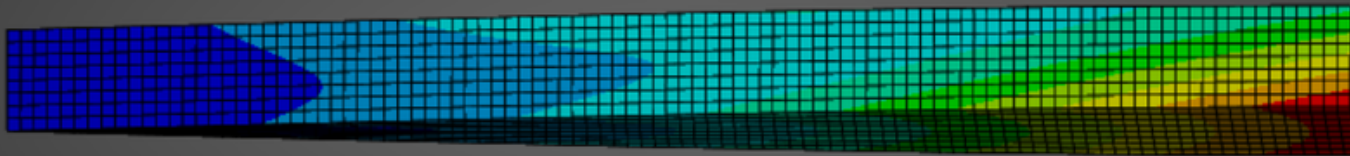
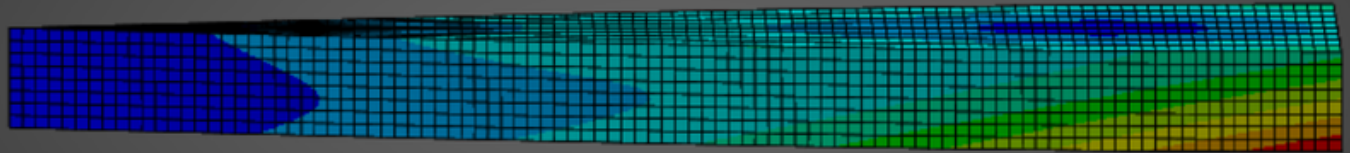
2408.7Hz



We compare the structural natural frequencies against the excitation frequency (200 Hz)

Mode 6

2643.9 Hz



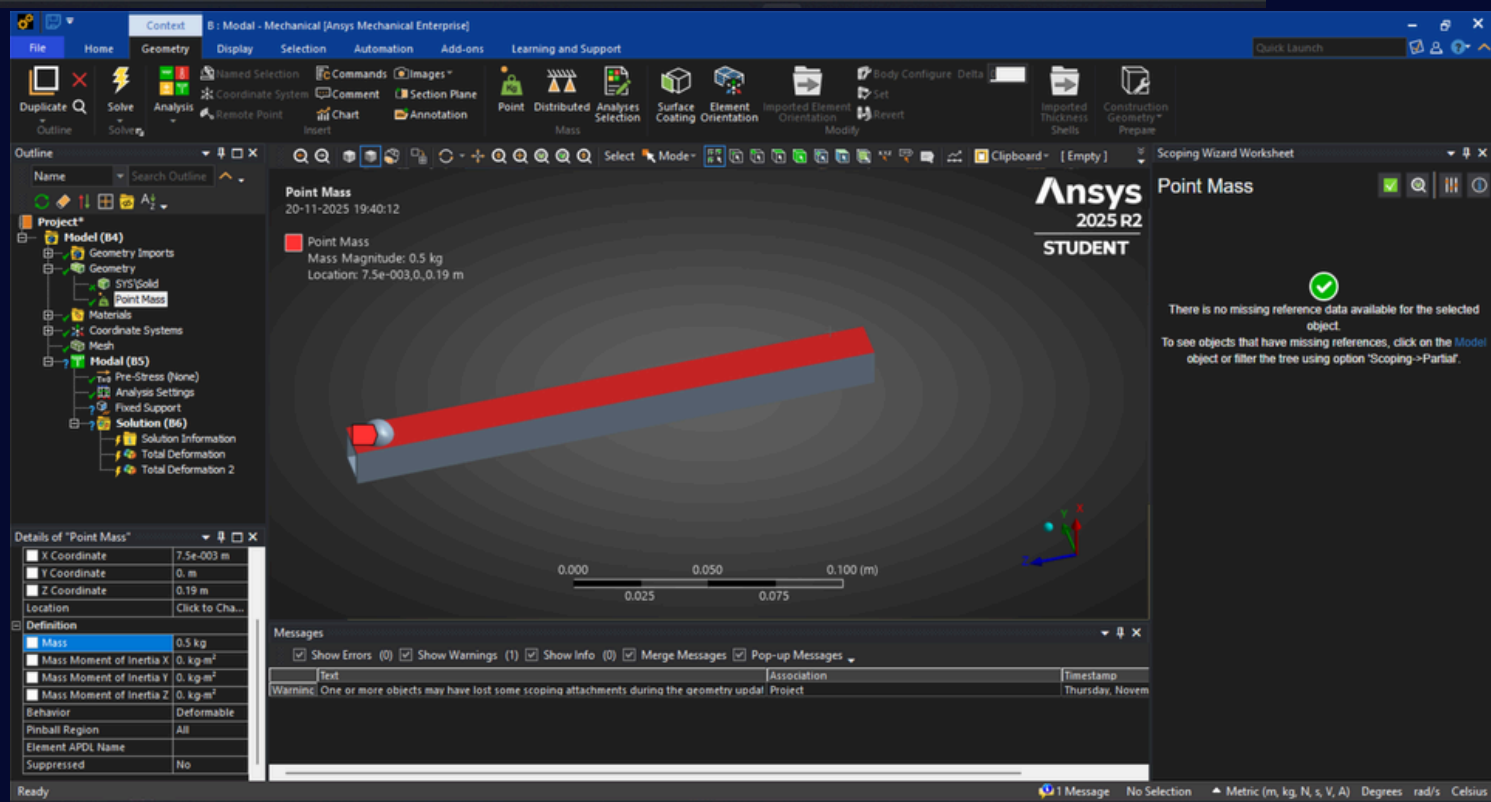
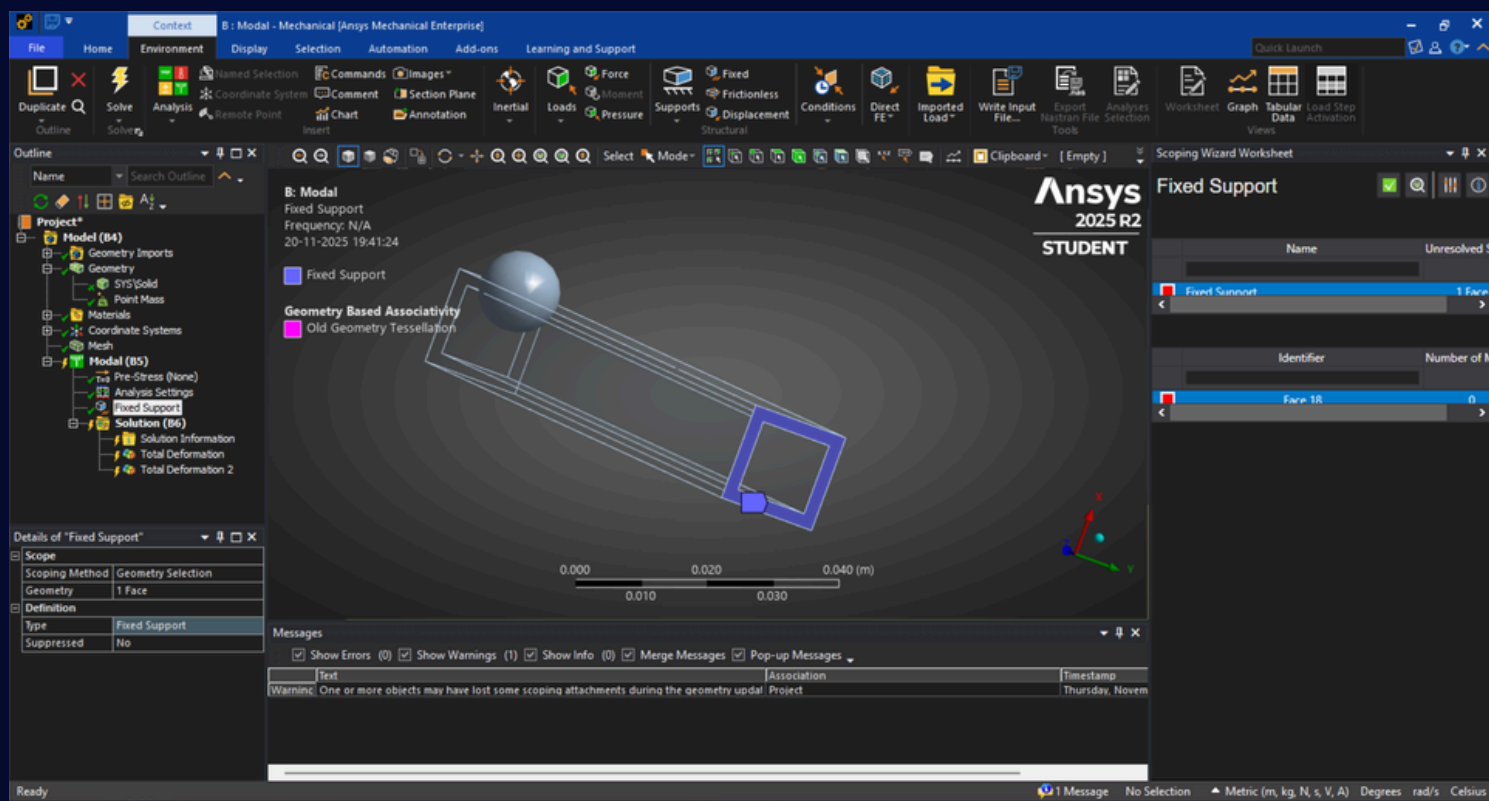
Thank You

The background features a dark blue field with several concentric circles in shades of purple, blue, and pink. Two bright, multi-pointed starburst graphics in a teal-to-blue gradient are positioned on either side of the 'Thank You' text.

Conclusion

DESIGN PASS

The Finite Element Analysis confirms that the drone arm design is structurally sound for dynamic loads at 12000 RPM. No resonance will occur at the target operating speed.



PHASE 2: MESH & BOUNDARY CONDITIONS

Accurate simulation requires careful meshing and realistic boundary conditions.



Mesh Application

Apply a standard mesh (e.g., 2mm element size). Vibration analysis demands a fine mesh for accuracy.



Fixed Support

Apply a fixed support to the base face of the arm, simulating its attachment to the drone body.

A high-quality mesh is paramount as vibration results are highly sensitive to element distribution and size.

